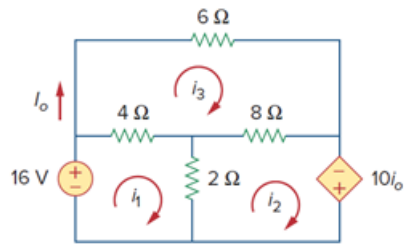


Problem

Using mesh analysis, find I_o in the circuit of Fig. 3.21.

Figure 3.21:



Step-by-step solution

Step 1 of 4

Draw the following circuit diagram shown in Figure 1.

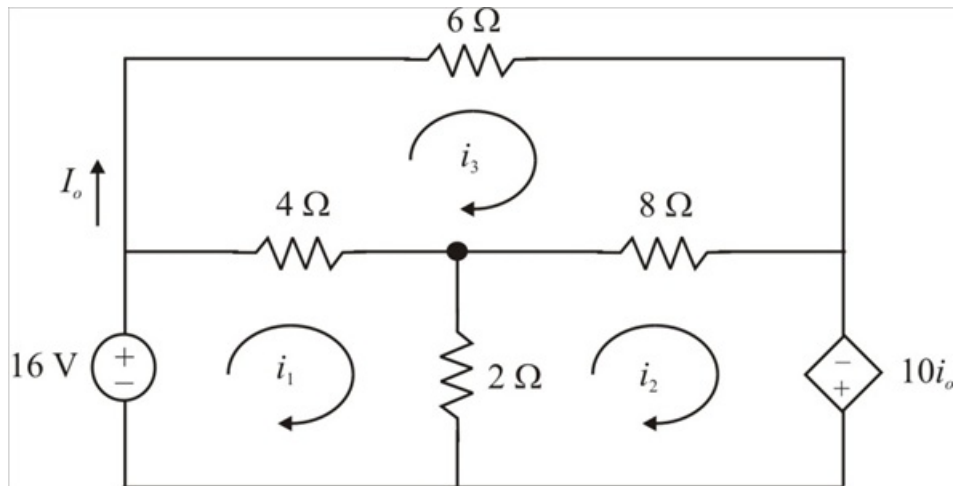


Figure 1

Step 2 of 4

Applying Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero.

For mesh 1, Kirchhoff's voltage law gives,

$$-16 + 4(i_1 - i_3) + 2(i_1 - i_2) = 0$$

$$-16 + 4i_1 - 4i_3 + 2i_1 - 2i_2 = 0$$

$$6i_1 - 2i_2 - 4i_3 = 16$$

$$3i_1 - i_2 - 2i_3 = 8$$

Therefore $3i_1 - i_2 - 2i_3 = 8$ (1)

For mesh 2, Kirchhoff's voltage law gives,

$$2(i_2 - i_1) + 8(i_2 - i_3) - 10i_o = 0$$

$$2i_2 - 2i_1 + 8i_2 - 8i_3 - 10i_o = 0$$

$$-2i_1 + 10i_2 - 8i_3 - 10i_o = 0$$

But, $i_o = i_3$

$$-2i_1 + 10i_2 - 8i_3 - 10i_3 = 0$$

$$-2i_1 + 10i_2 - 18i_3 = 0$$

$$-i_1 + 5i_2 - 9i_3 = 0$$

Therefore $-i_1 + 5i_2 - 9i_3 = 0$ (2)

Step 3 of 4

For mesh 3, Kirchhoff's voltage law gives,

$$4(i_3 - i_1) + 6i_3 + 8(i_3 - i_2) = 0$$

$$4i_3 - 4i_1 + 6i_3 + 8i_3 - 8i_2 = 0$$

$$-4i_1 - 8i_2 + 18i_3 = 0$$

$$-2i_1 - 4i_2 + 9i_3 = 0$$

Therefore $-2i_1 - 4i_2 + 9i_3 = 0$ (3)

Equation (1), (2) and (3) can be cast in matrix form as follows:

$$\begin{pmatrix} 3 & -1 & -2 \\ -1 & 5 & -9 \\ -2 & -4 & 9 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \\ 0 \end{pmatrix}$$

By using Cramer's rule, determinant, Δ can be calculated as follows:

$$\begin{aligned} \Delta &= \begin{vmatrix} 3 & -1 & -2 \\ -1 & 5 & -9 \\ -2 & -4 & 9 \end{vmatrix} \\ &= 3(45 - 36) + 1(-9 - 8) - 2(9 + 10) \\ &= 27 - 17 - 38 \\ &= -28 \end{aligned}$$

By using Cramer's rule, determinant, Δ_1 can be calculated as follows:

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} 8 & -1 & -2 \\ 0 & 5 & -9 \\ 0 & -4 & 9 \end{vmatrix} \\ &= 8(45 - 36) - 0(72 - 8) + 0(9 + 10) \\ &= 72 - 0 - 0 \\ &= 72 \end{aligned}$$

Step 4 of 4

By using Cramer's rule, determinant, Δ_2 can be calculated as follows:

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 3 & 8 & -2 \\ -1 & 0 & -9 \\ -2 & 0 & 9 \end{vmatrix} \\ &= 3(0+0)+1(72+0)-2(-72+0) \\ &= 0+72+144 \\ &= 216\end{aligned}$$

By using Cramer's rule, determinant, Δ_3 can be calculated as follows:

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} 3 & -1 & 8 \\ -1 & 5 & 0 \\ -2 & -4 & 0 \end{vmatrix} \\ &= 3(0+0)+1(0+32)-2(0-40) \\ &= 0+32+80 \\ &= 112\end{aligned}$$

Thus, since $i_o = i_3$, the current can be calculated as:

$$\begin{aligned}i_3 &= \frac{\Delta_3}{\Delta} \\ &= -\frac{112}{28} \\ &= -4 \text{ A}\end{aligned}$$

Therefore the current i_3 is -4 A.